

Abstract

Chatham County is highly vulnerable to flood hazards, and has seen frequent flooding of infrastructures, roads and properties threatening public safety. Due to the low elevation, relatively flat terrain, proximity to the coast, high tides, and abundant rivers surrounding and traversing the county, flood hazards in Chatham County include (1) coastal floods and nuisance driven by high tides, storm surge and sea-level rise; (2) flash floods as a result of intense precipitation and surface water runoff; and (3) riverine floods caused by the rise of water level. Chatham County acutely is vulnerable to compound flood events. Due to more frequent extreme climate events and sea-level rise projected with climate change throughout the 21st century and beyond, compound flood hazards must be considered for risk assessment, forecasting and mitigation plans for storm water management. Unfortunately, very few if any such studies exist for coastal regions of the southeast. The project aims to generate new data and develop new models, tools and maps to assess compound coastal flooding risks for storm water management in Chatham County. Project products will consider the most accurate and up-to-date data from both inland urban hydrology, hydraulics in storm water management systems (drainage, pipes, canals and rivers), tides and sea level rise, high-resolution digital data of urban development. Eventually, we aim to generate up-to-date and forecasting flood maps that identify dynamic “hot spots” (e.g., pipelines, roads and areas) with the highest risks of flooding during heavy rainfall associated with hurricanes and tropical storms, and to highlight those components of the storm water management and infrastructures whose maintenance, repair, and/or upgrade are most critical to flood risk mitigation in Chatham County

Motivation and Introduction

Floods in coastal regions caused by sea level rise (SLR) and extreme weather events have become the most common and costly natural disasters in the world. In the U.S., floods caused by climate change are the top-priority national security challenge for emergency management [2]. Several flood models have been developed to estimate flood-hazard exposure in coastal regions to account for the impacts of SLR, tides, and storms [1]. For example, the hydrodynamic Sea, Lake, and Overland Surges from Hurricanes (SLOSH) model was developed to simulate storm surge from tropical cyclones and the Coastal Storm Modeling system (CosMoS) recently developed by U.S. Geological Survey (USGS) was applied to urbanized areas in California [3]. One of the main assumptions of these models is that coastal impacts such as storm surges are the dominant flood driving factors, while floods in urbanized coastal regions are driven by urban hydrology, river and open channel hydraulics, coastal impacts, and climate change. Chatham County is a typical case for its high-elevation urban areas and low-lying wetlands and coastlines. Therefore, most of its flood events were caused by heavy rainfall and drainage system failure and decreased infiltration capacity caused by SLR, river water level rise and wetland inundation. However, such a model integrating urban hydrology, drainage systems, river hydraulics, and coastal impacts has not yet been available for coastal Georgia. This research aims to develop a comprehensive urban-hydrology-hydraulics model to assess coastal flooding while integrating multiple flooding drivers, such as SLR, river stage data, extreme weather events, and intensive urbanization. Specific questions that will be addressed include:

- (1) How will the main flood drivers change and interact in future scenarios?
- (2) How will SLR and climate change the exposure risk and vulnerability map of urban infrastructure?
- (3) How will SLR and climate change affect the capacity of current urban drainage systems, including infiltration and river discharge?
- (4) In future scenarios of SLR and climate change, how to improve urban planning and drainage systems to adapt to and alleviate compounded flooding to enhance resilience?

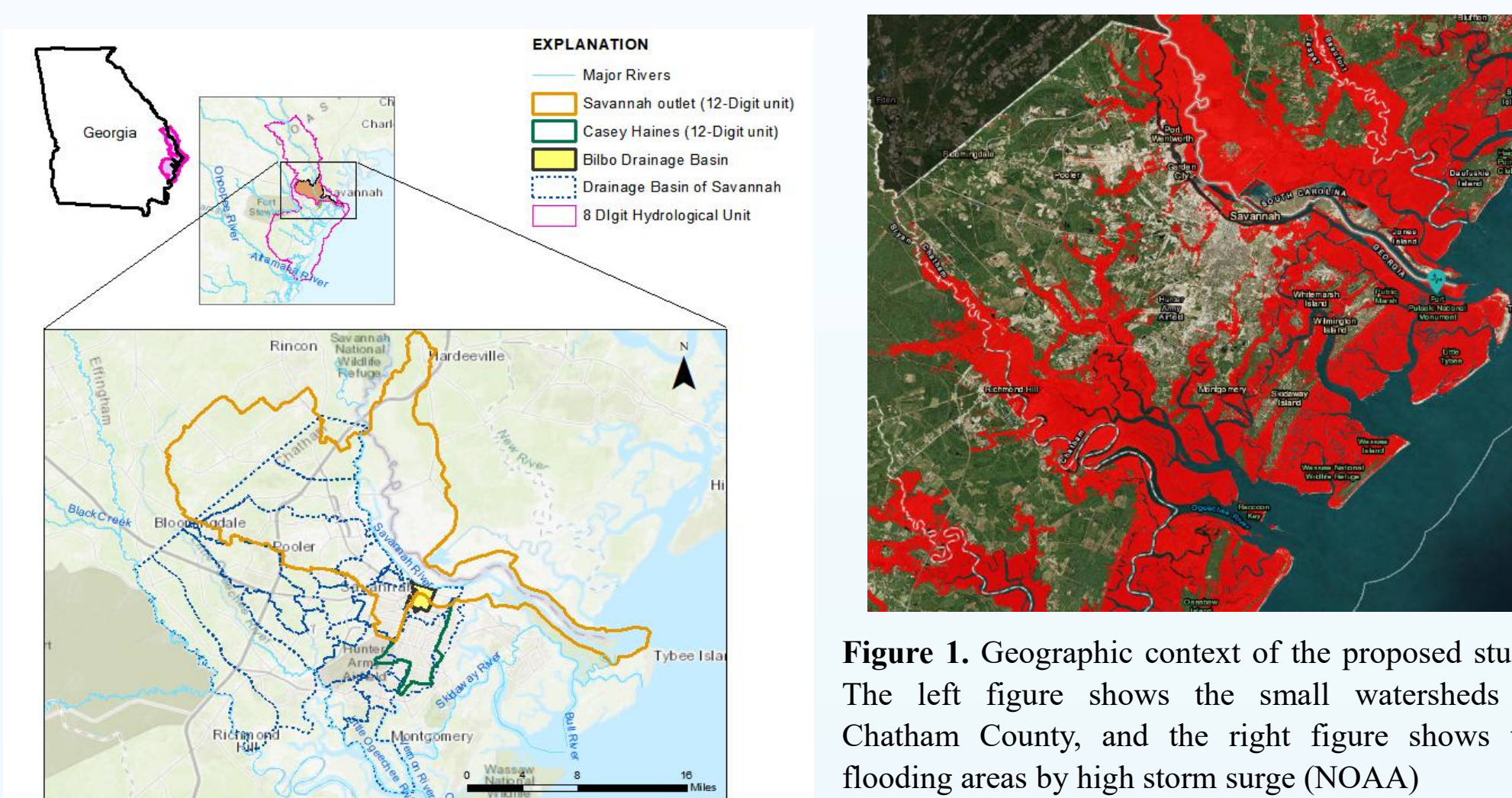
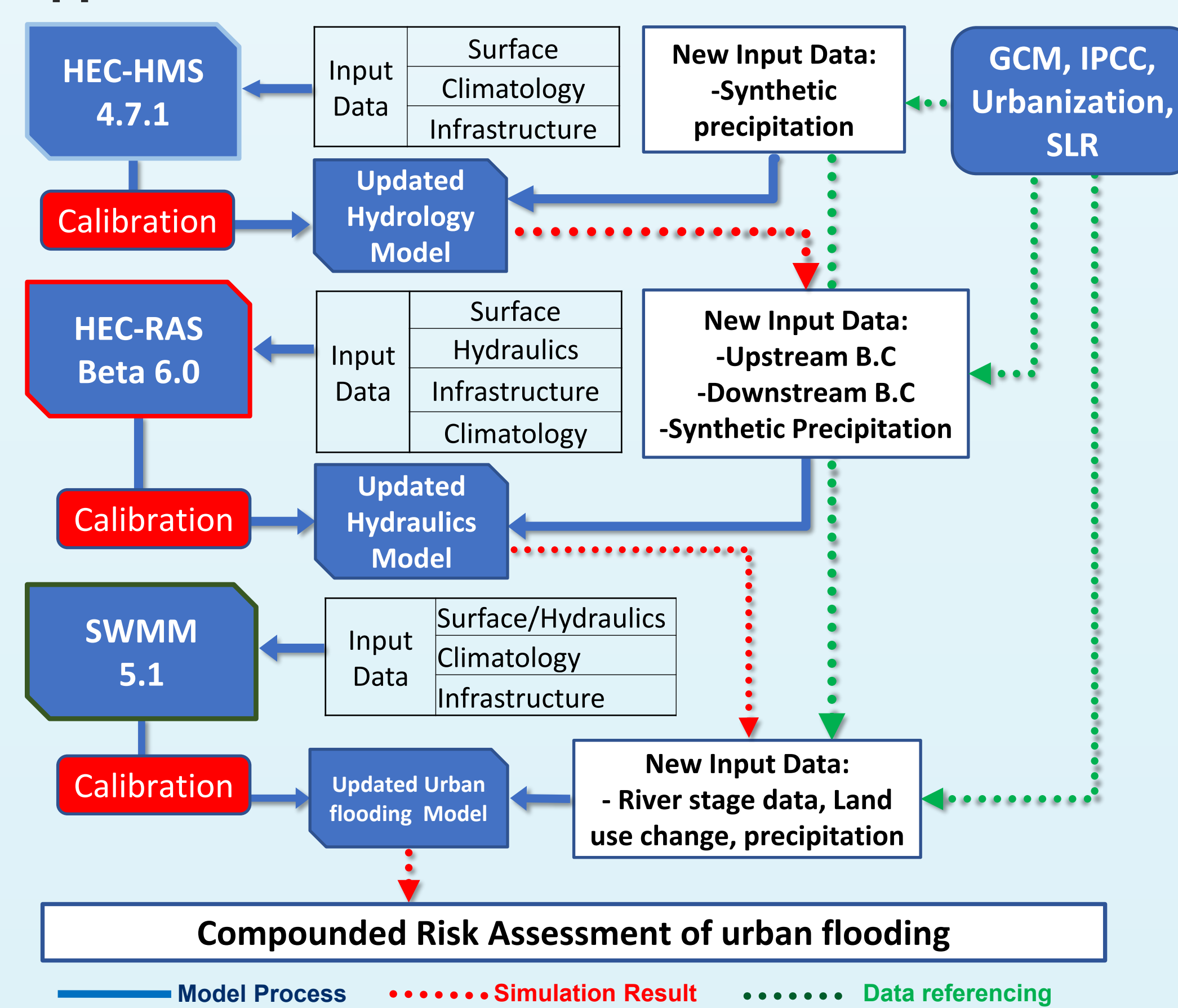


Figure 1. Geographic context of the proposed study. The left figure shows the small watersheds in Chatham County, and the right figure shows the flooding areas by high storm surge (NOAA)

Method

Modeling Frame:

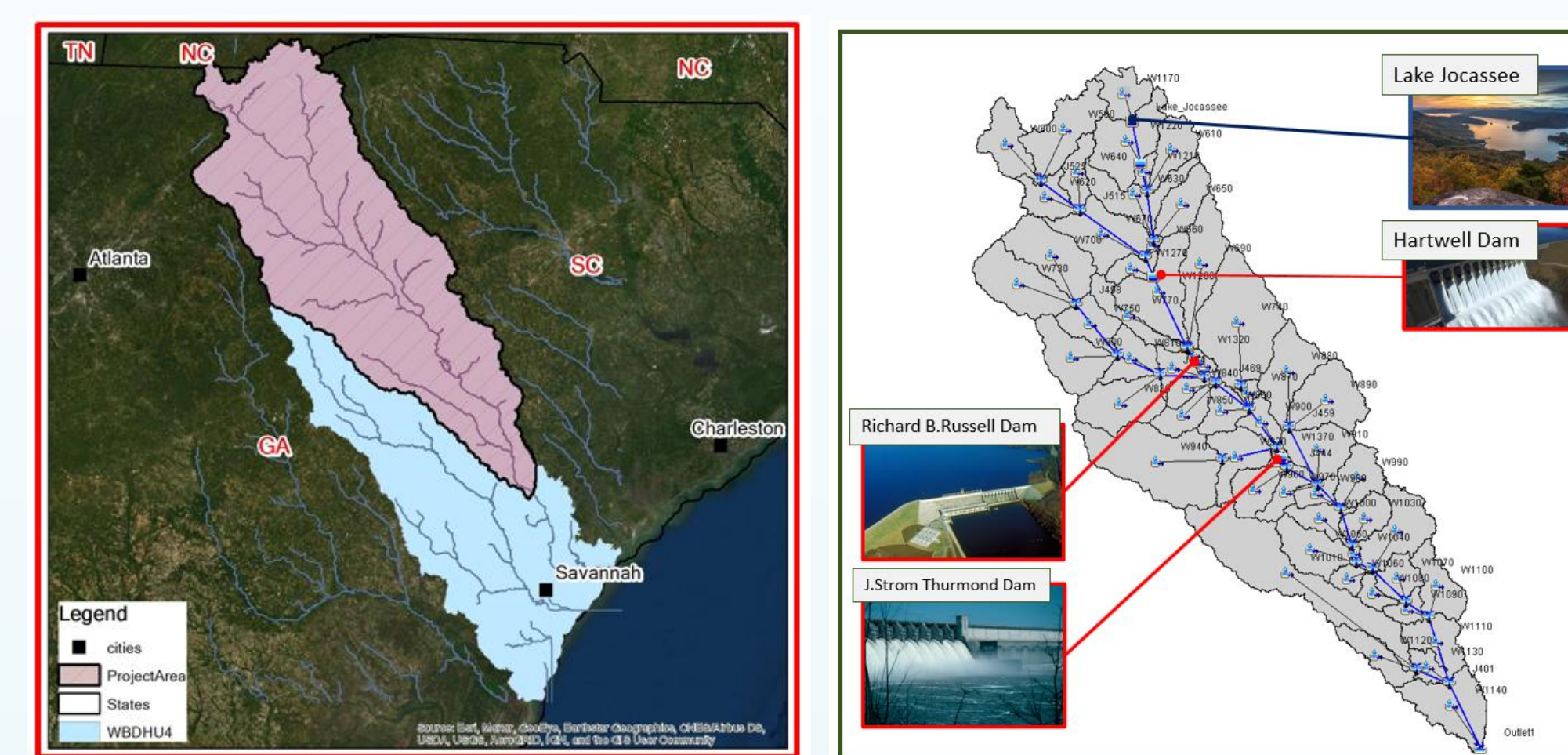
The model framework is to integrate three types of models which include multiple disciplines, (1) hydrology model using HECHMS [4] (2) hydraulics models using HECRAS [5], (3) urban-hydrology-hydraulic models to assess and predict compounded flood impacts using SWMM [6]



As the model diagram presents the integrated model can produce compounded flood risk assessment considering flood drivers such as geography, precipitation, land use, coastal impact all combined with climate change. The hydrology model provides flowrate with river stage data to hydraulics model. The hydrological model products take a role as boundary conditions for hydraulics model while another boundary condition comes from coastal parts. Urban hydraulics- hydrology model will be provided its boundary conditions from two models above. Then the constructed model will be calibrated using data sets from many agencies (NOAA, USGS, CEMA, City of Savannah, Georgia Emergency Plan)

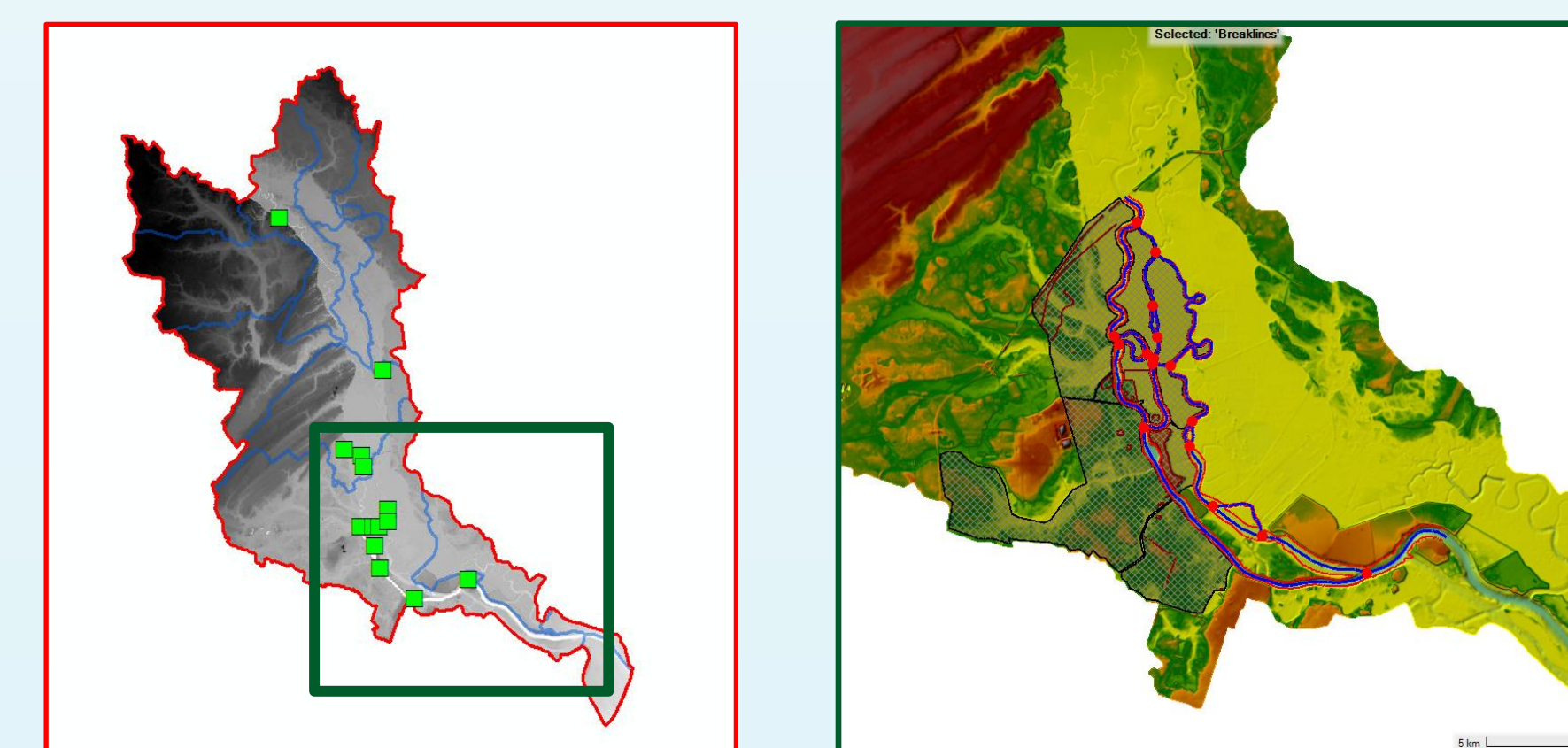
Task1-1. Hydrology model:

Northside of the Chatham county is surrounded by Savannah River which is one of the major boundary conditions for the project. A diagram above show hydrological model domain and HECHMS model for Savannah River. The Savannah river flow is controlled by three major reservoirs of Richard B. Russell Dam, J.Strom Thurmond Dam, and Hartwell Dam. The input data set are precipitation, surface conditions, reservoir control rules, elevation. The output of the hydrology model is flowrate to an outlet of the hydrology model which will be an input data for inlet of river hydraulics model.



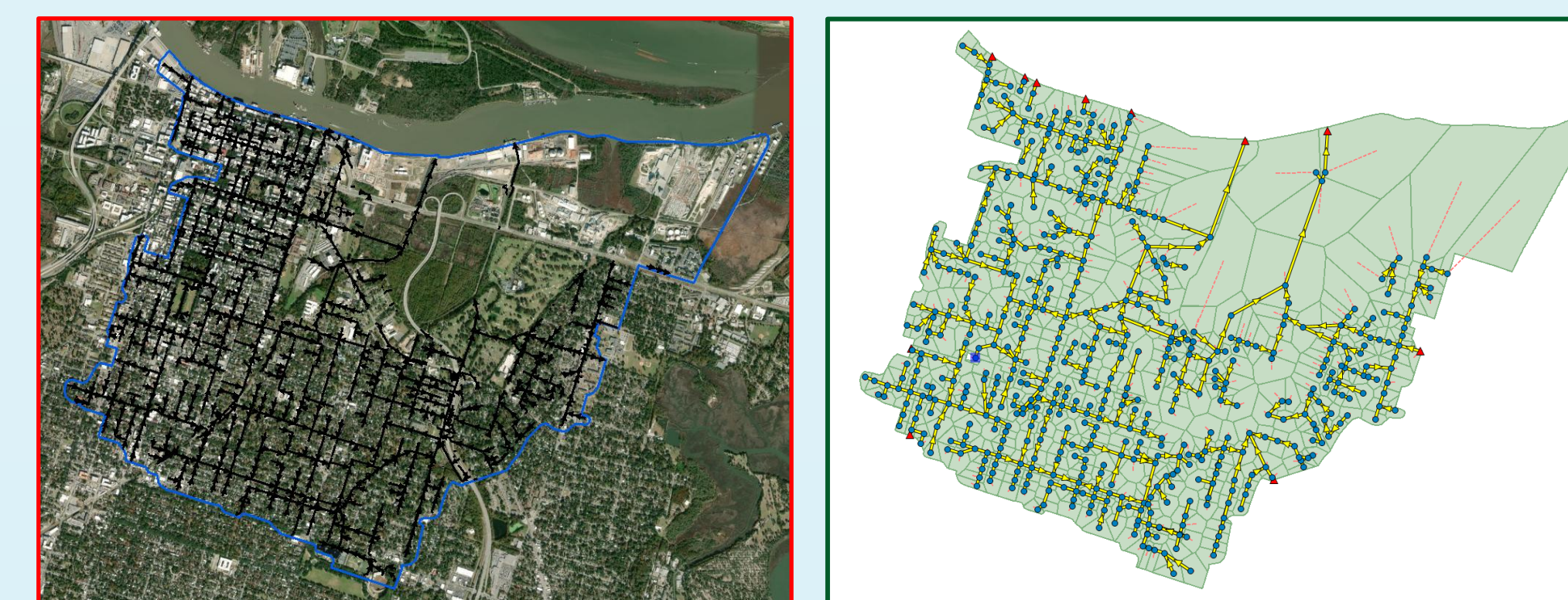
Task 1-2. Hydraulics model:

Hydraulics model under construction is built on HECRAS Beta 6.0 Beat. Advanced from previous HECRAS 5.0, the current beta version can perform hydrological modeling for subcatchment with 2D method. The hydraulics model will provide boundary conditions for urban hydraulics model to assess stormwater management system combined with climate change, SLR, intensified urbanization. Coastal impacts will be considered as one of the major boundary conditions and drivers to coastal flooding. The hydraulics model will simulate river stage data of Savannah River and flood plain inundation due to flooding.



Task 1-3: Urban Hydrology-hydraulics model

A diagram below shows a pilot model for urban hydraulics-hydrology model for city of Savannah. The routing of urban hydraulics model transports surface runoff from the urban hydrology model through a network of pipes, channels, storage ponds and pumps. Principal components of a hydraulics system are the pipeline network, inlets to pipes that define the sub-catchment runoff to be discharged, and outfalls, which are integrated with hydrology, storm surge, SLR by quantifying the degradation of discharge capacity of the hydraulic system during extreme weather events. HECRAS and SWMM will be combined or coupled together to conduct accurate model of the stormwater management system.

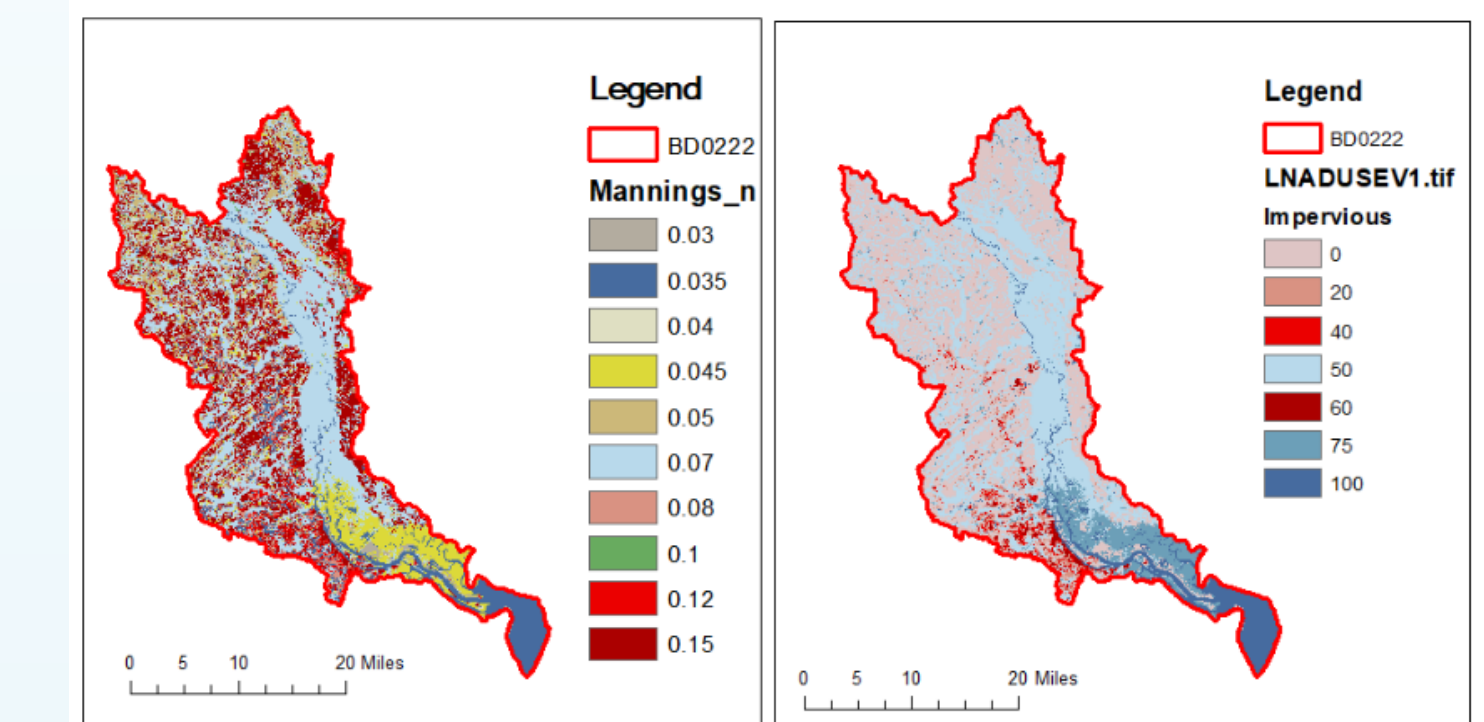


Task 2: Model Validation and sensitivity with uncertainty analysis

Historical data of flooding events will be used to calibrate the developed model. Sensitivity analysis will be performed to understand the importance of model inputs and obtain first-order results of flooding risks in terms of storm water management systems and driving factors. Statistical analysis based on uncertainties will be used to generate risk and vulnerability maps.

Task 3: Adaption and mitigation strategies in future scenarios

In this task, future scenarios of SLR and climate change will be generated, including four trends of static SLR up to 2 m until 2100 and extreme weather events (5 years, 20 years, 50 years and 100 years). Another consideration human activities to trigger increased flooding risk such as intensified urbanization, corresponding land use. Based on the results from Task 2, we will examine the performance of potential strategies for improve the storm water management systems.



Example of input data
need to be modified
along future scenarios
(Physical surface
conditions)

Deliverable Results

The project can help identify flood risk and the major factors that drive flooding and has the potential to provide technical support to stake holders. The research can help coastal communities increase their understanding of changing conditions and identify risks so that they become more resilient to coastal flooding, economic disruptions, and other threats to well-being of community.

- 1) **Compounded Flood Risk assessment**
 - Identify major coastal flooding driver(s) in Chatham county
 - Provides toolkit for flood analysis, and improvement of stormwater management system
- 2) **Technical support for land use plan and adaptation/mitigation plan to coastal flooding**
 - Assessment of adaptation plan to changing conditions
 - Assessment of current/ future land use
- 3) **Flood risk mapping public/ commercial use**
 - Generate an easy and accessible map of flooding risk to public
 - Enhance awareness of coastal flooding to public/ local economy

References

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- [4] U.S. Army Corps of Engineers. (2000b). "HEC-HMS Technical Reference Manual," Hydrologic Engineering Center, Davis, C
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Acknowledgements

The project is supported by Sea Grant of Georgia (University of Georgia and NOAA)